

Instructions:

1. Attempt both sections in separate answer books.
2. Figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION-I

Q.1[A] Do as directed: **[09]**

1. Full form of JRE is ____.
2. ____ is the default *package* included in all java programs.
a. lang b. java.lang c. java.language d. language
3. In ____ object relationship, both the entities can survive individually.
a. Composition b. Association c. Aggregation d. Abstraction
4. Prefix ____ is used to indicate hexadecimal number.
a. 0 b. hex c. x d. 0x
5. ____ is use to call a *parameterized constructor* from another constructor.
a. this(values); b. this.constructor(values); c. this() d. this.classname(values);
6. ____ is a valid identifier.
a. :abc b. #abc c. \$abc d. &abc
7. ____ method will be called just before the object of a class is claimed to be recycled.
a. destroy() b. finalize() c. final() d. finally()
8. ____ is collection of only *abstract methods*.
a. abstract class b. parent class c. interface d. final class
9. Size of *byte* datatype is ____ bits.
a. 4 b. 8 c. 16 d. 32

[B] Explain *command line arguments* in Java? **[02]**

Q.2 Answer the following questions. **[12]**

[A] Explain *static* keyword with example.

[B] List and explain *pillars of Object Oriented Programming Language*.

[C] Differentiate:

- a. C++ and Java
- b. *Method* and *constructor*

OR

Q.2 Answer the following questions. **[12]**

[A] Explain *final* keyword with example.

[B] Explain the working of *JVM* with diagram.

[C] Differentiate:

- a. *Abstract class* and *Interface*
- b. *Method overriding* and *Method overloading*

Q.3 Answer the following questions. [Any three] **[12]**

[A] What is *constructor*? Explain types of *constructor* with example.

[B] List and explain features of Java.

[C] List and explain types of *inheritance* in Java with suitable example.

[D] List and explain *operators* in Java with example.

[E] Explain *abstract* keyword with example.

SECTION-II

Q.4[A] Do as directed:

[09]

1. equals () method of String class compares _____ of two strings.
a. contents b. reference c. relation d. both A and B
2. _____ is the default layout for *Frame*.
a. FlowLayout b. BorderLayout c. GridLayout d. BoxLayout
3. Which of the following is a valid access modifier?
a. static b. final c. protected d. synchronized
4. Which of the following is **not** a Thread priority?
a. AVG_PRIORITY b. NORM_PRIORITY c. MAX_PRIORITY d. MIN_PRIORITY
5. Which of the following is **not** a valid String concatenation?
a. str1.concat(str2); b. str1.add(str2); c. str1 + str2; d. " ".concat(str2)
6. _____ thread begins running immediately, when a java program starts up.
a. this b. current c. main d. Thread1
7. *Wrapper class* objects are stored in _____ memory.
a. volatile b. constant pool c. stack d. heap
8. _____ method pauses currently executing thread for specified milliseconds.
a. wait() b. yield() c. join() d. sleep()
9. _____ package provides classes for all components of GUI.
a. java.lang b. java.util c. java.gui d. java.awt

[B] Differentiate: Throw and Throws

[02]

Q.5 Answer the following questions.

[12]

- [A]** How to create a thread by extending *Thread class*? Explain it with example.
- [B]** Explain *BorderLayout* with example.
- [C]** Explain different ways to import classes from a *package*.

OR

Q.5 Answer the following questions.

[12]

- [A]** How to create a thread by implementing *Runnable interface*? Explain it with example.
- [B]** Explain *FlowLayout* with example.
- [C]** What is *user-defined package*? Write the steps to create user-defined package.

Q.6 Answer the following questions. [Any three]

[12]

- [A]** Draw and explain life cycle of *Thread*.
- [B]** List and explain the keywords used in *exception handling* in Java.
- [C]** Explain *TextField* control with its constructor and methods.
- [D]** Explain *autoboxing* and *unboxing* in java with example.
- [E]** Write a java program that adds label and textbox in a *frame* as shown below.

